CST8216 Week 2

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AhaSlides

What does voltage measure in an electrical circuit?

0	0	0	0
✗ The number of electrons in the circuit	✗ The speed of electron flow	✓ The potential difference between two points	✗ The resistance between two points

1

CST8216 Week 2

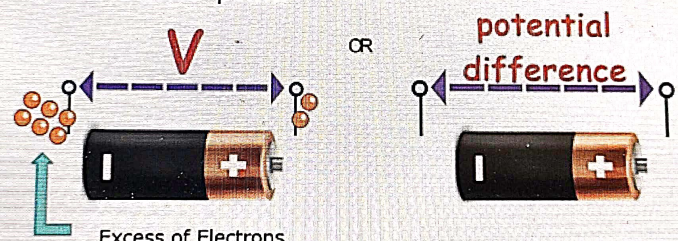
My Answers | All Slides

AhaSlides

Voltage (symbol: E, or V; unit: volt or V) – Definition and Characteristics

Voltage is defined as the electrical potential difference between two points

Voltage DOESN'T exist at a point by itself – it is always determined with respect to some other point.



2

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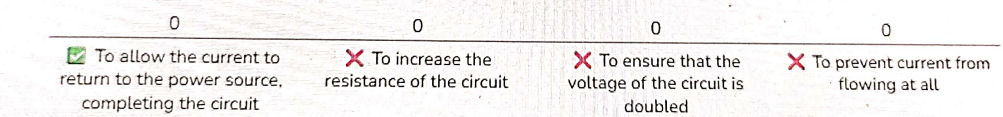
Why is a closed loop necessary for the flow of current in an electrical circuit?



◀ 3 ▶

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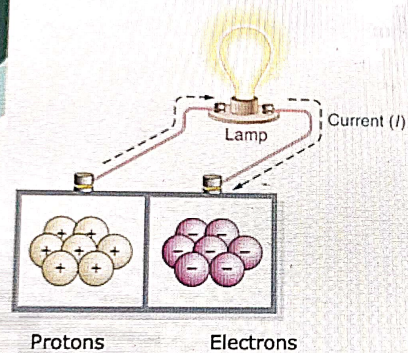
How all of this works – the Pictorial Diagram



The path between the two terminals of the battery is now a "**closed loop**"; current is generated because of the applied Voltage.

The rate at which the current flows is determined by the resistance of the Lamp, *given that the voltage is constant*. .

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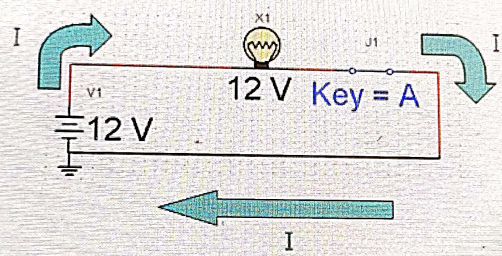
How all of this works – the Schematic Diagram



because the circuit is a CLOSED loop...

5

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What is the role of the potential difference in a battery?

✗ It indicates the physical distance between the battery's terminals.	✗ It prevents the battery from discharging.	✓ It drives the flow of electrons when the circuit is closed.	✗ It measures the resistance of the battery.
0	0	0	0

6

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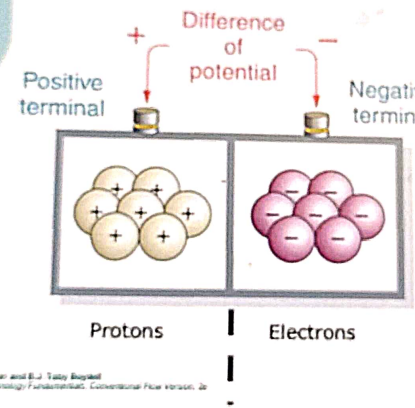
How all of this works – the Pictorial Diagram



A battery "charged" to 12 Volts has a potential difference of 12 Volts between its positive and negative terminals.

At this point, however, there is no flow of "charge", because we have an "**open loop**".

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What happens if a resistor in a circuit exceeds its power rating?

0	0	0	0
✗ It will increase the overall resistance of the circuit.	✗ It will improve the efficiency of the circuit.	☑ It may burn out or get damaged.	✗ It will decrease the voltage drop across it.

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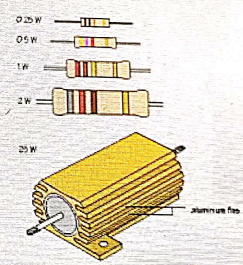
Resistance (symbol: R; unit: ohm or Ω) – Definition and Characteristics

Resistance is defined as the opposition to current.

A resistor is constructed to have a specific amount of resistance to current flow.



- If the current is excessive, the resistor will burn out because its power rating has been exceeded.
- In addition to having a specific resistor value, resistors also come in various sizes. The larger the cross-sectional area, the more heat can be dissipated by the resistors

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What determines the rate at which current flows through a lamp in a closed electrical circuit?

0	0	0	0
✗ The color of the lamp	✓ The voltage provided by the battery and the resistance of the lamp	✗ The size of the battery	✗ The length of the wires in the circuit

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What happens to the brightness of a lamp if the resistance in the circuit is increased?

0	0	0	0
✗ The lamp glows brighter.	✗ The lamp glows with the same brightness.	✓ The lamp glows less brightly.	✗ The lamp's color changes.

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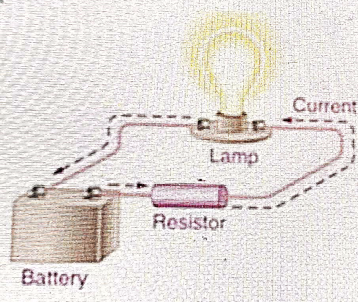
What is the effect of adding Resistance?



Adding Resistance to this circuit will limit the amount of current that can flow in the circuit.

Resistance is the opposition to current. The Resistor in this circuit limits the amount of current flow that will be allowed to pass through the Lamp. As such, the Lamp will not glow as brightly as it would without the added Resistor.

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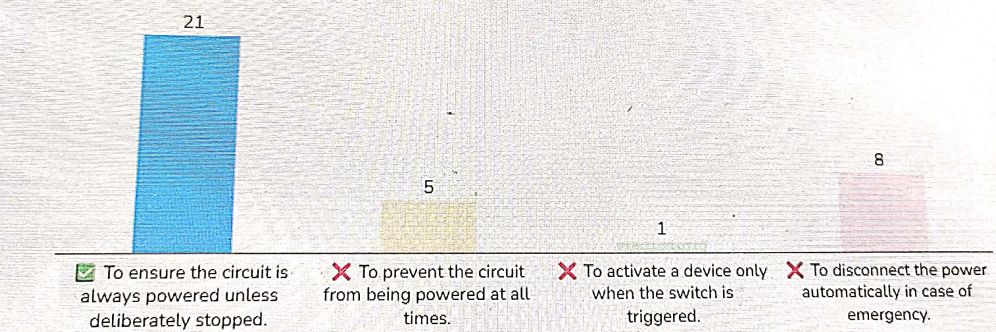
What is the primary use of a normally closed (NC) switch in safety applications?



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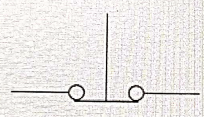
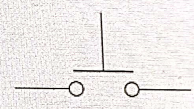
Normally closed (NC) and normally open (NO) switches.



(a) Normally closed (NC) switch



(b) Normally open (NO) switch

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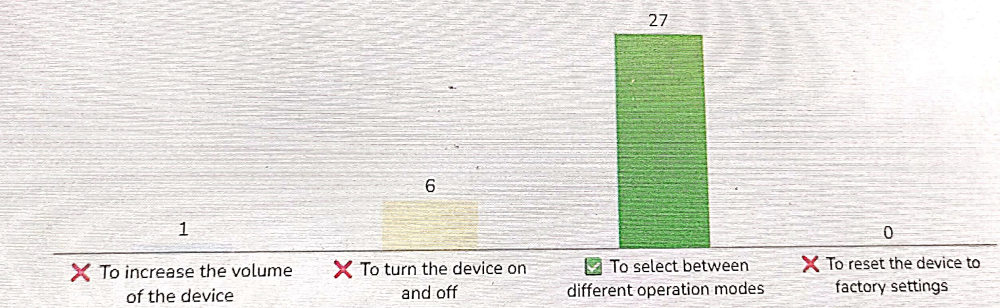
What is a common use of rotary switches in electronic devices?



16

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A rotary switch.

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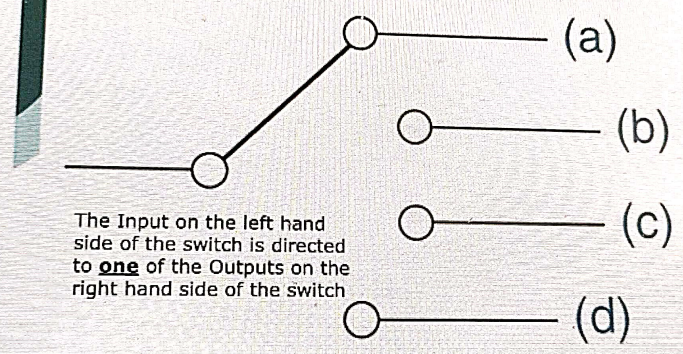

The Input on the left hand side of the switch is directed to **one** of the Outputs on the right hand side of the switch

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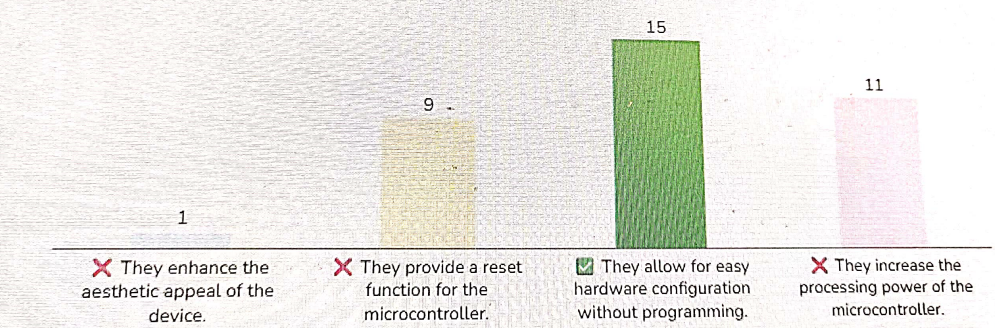
What is the primary advantage of using DIP switches in an embedded system?



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DIP switches.

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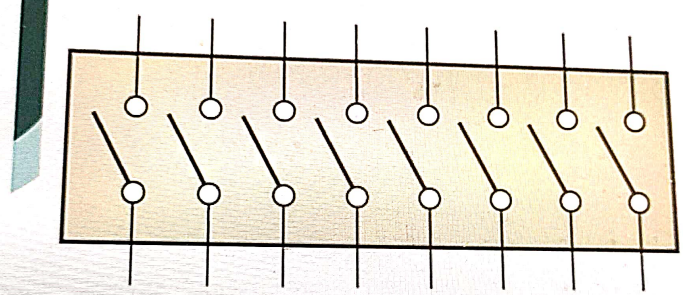

(b) Schematic symbol

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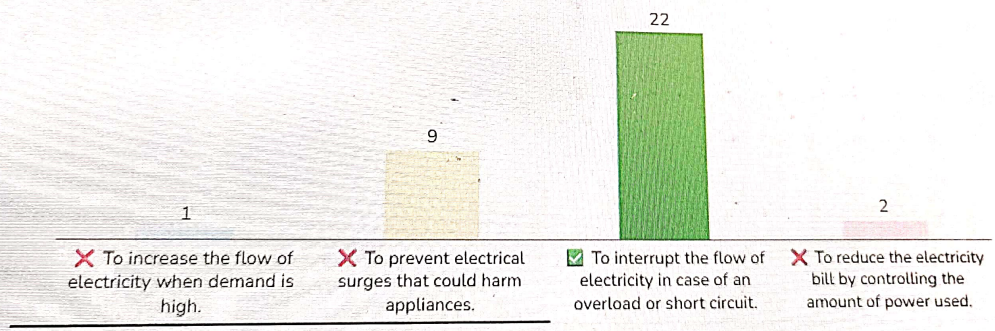
What is the primary function of a circuit breaker in an electrical circuit?



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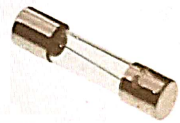
Circuit protection devices – Fuses and Circuit Breakers

- Excessive current flow in an electrical circuit may damage the circuit.
- Typical causes of increased circuit flow are low-resistance connections outside the normal path of current flow, called ***short circuits***.
- Fuses and circuit breakers provide protection from excessive current flow.



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familyhandyman.com

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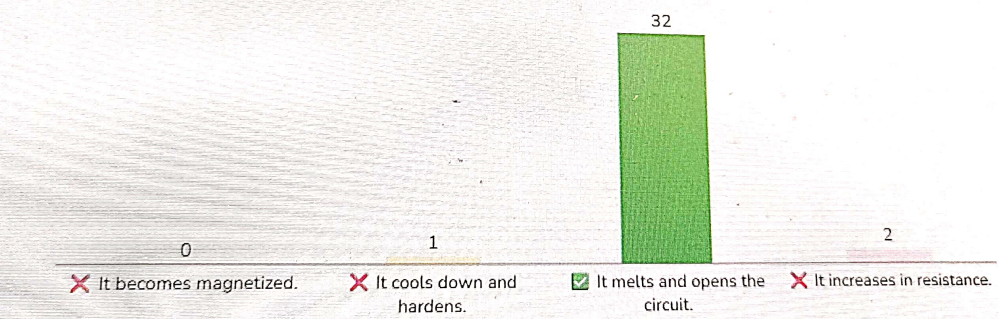
What happens to the resistive link inside a fuse when excessive current flows through it?



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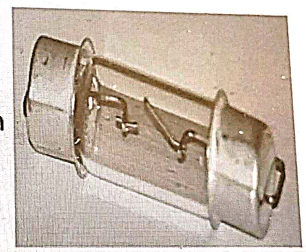
Fuses

- A resistive link inside the fuse heats up when current flows through it. Excessive current flow causes the link to burn, which stops the current flow and protects the circuit.
- A fuse that has burned open is said to be ***blown***.



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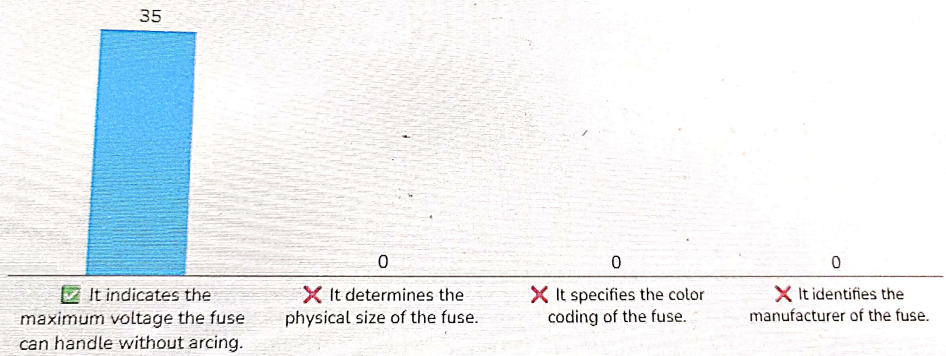
What is the purpose of the voltage rating on a fuse?



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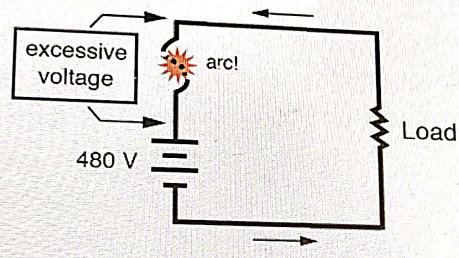
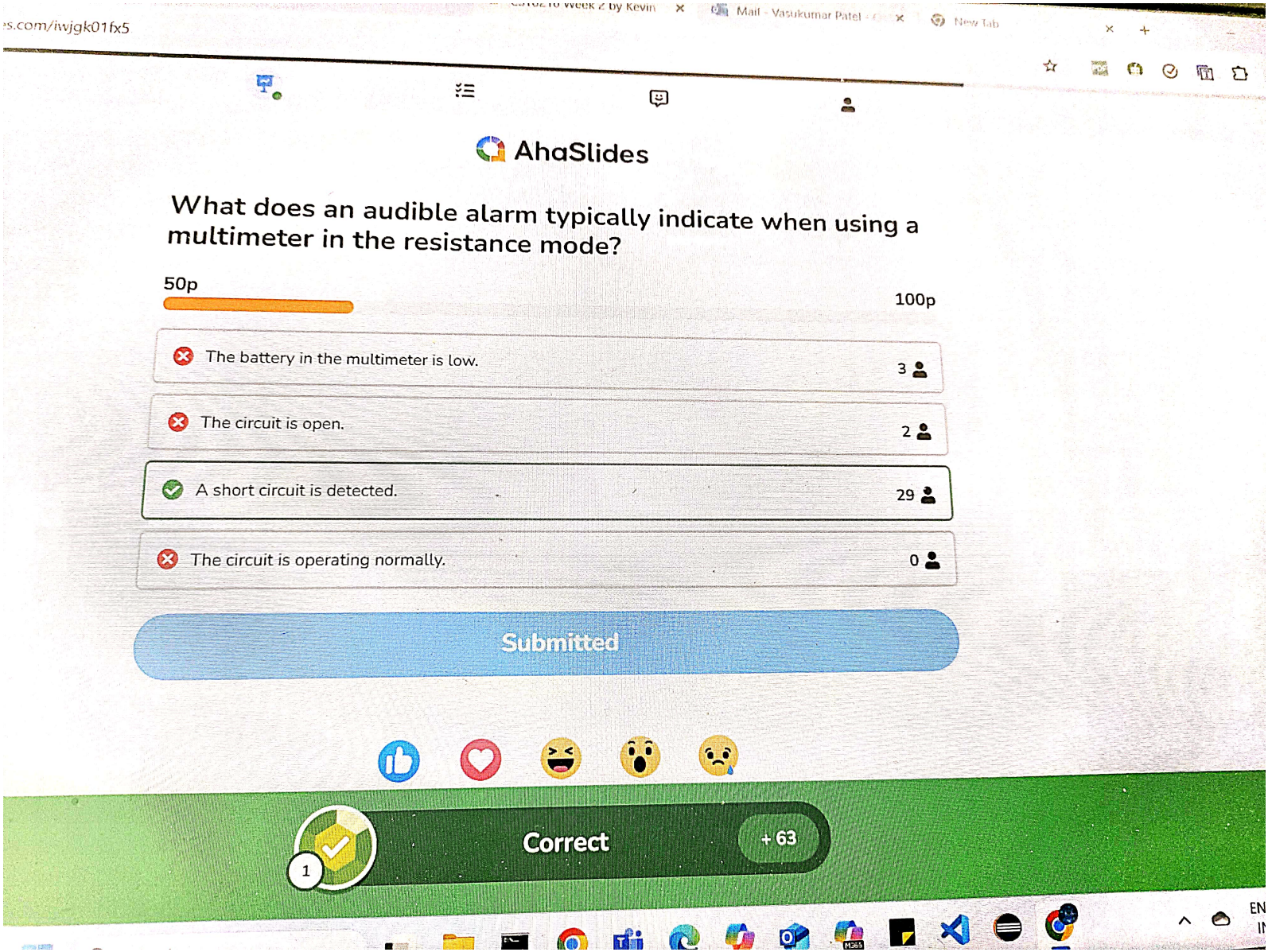


If the voltage across the blown fuse is high enough, a spark may jump the gap, allowing some current in the circuit.
THIS WOULD NOT BE GOOD!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AhaSlides
What does an audible alarm typically indicate when using a multimeter in the resistance mode?
50p
100p
The battery in the multimeter is low.
3
The circuit is open.
2
A short circuit is detected.
29
The circuit is operating normally.
0
Submitted
1
Correct
+ 63

les.com/wgk01fx5

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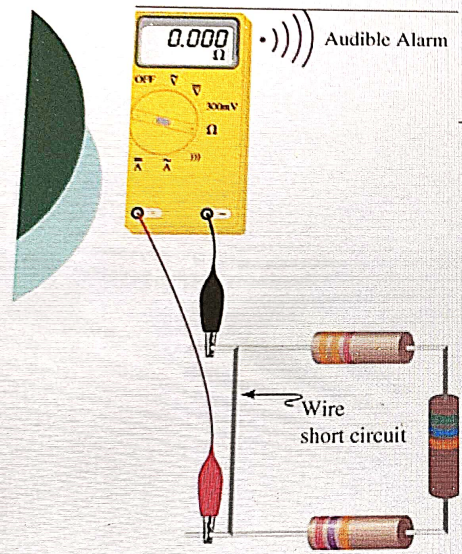
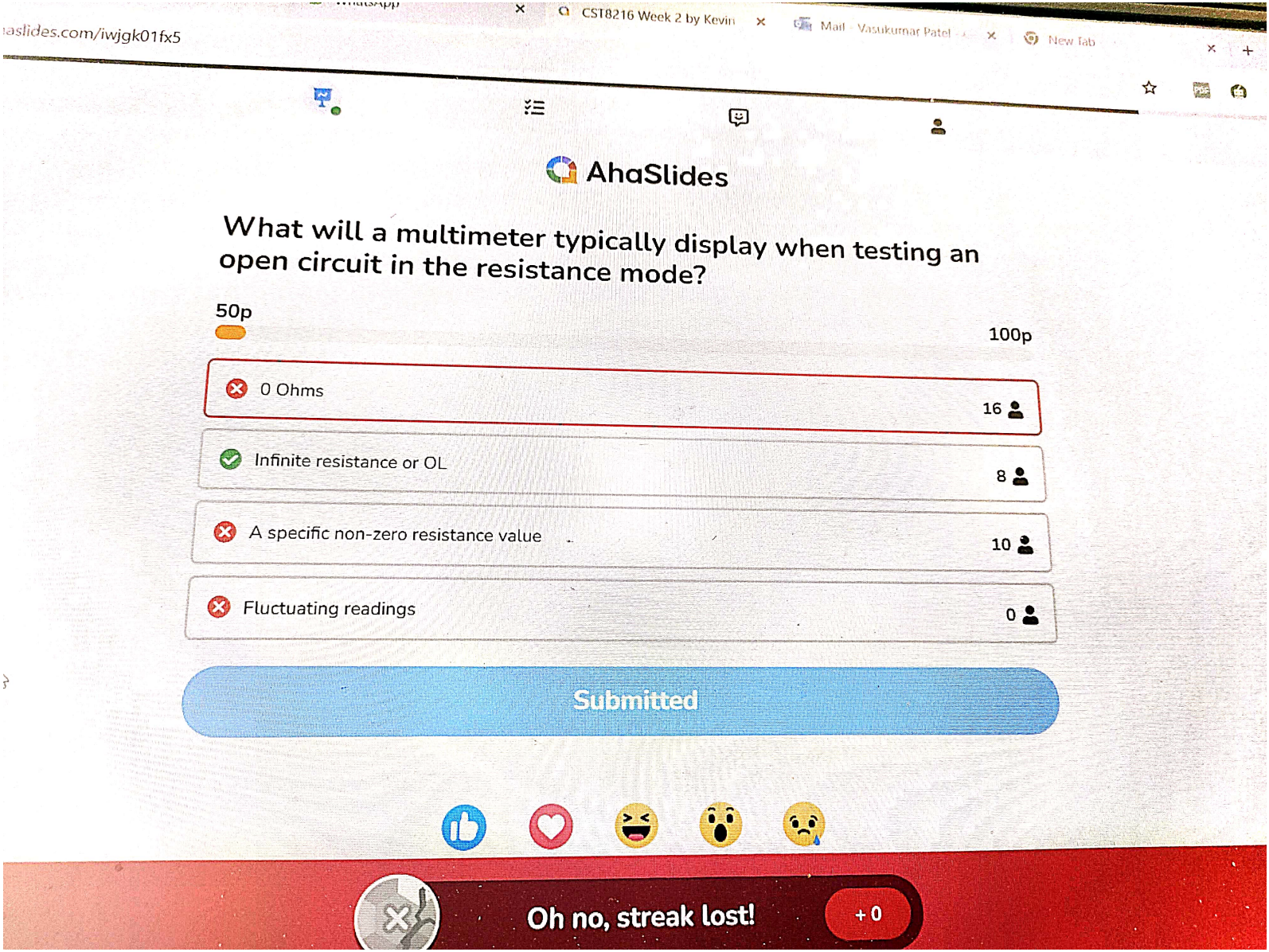
The Short Circuit…



(a) Short circuit

8 10 20 124 33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New Tab
AhaSlides
What will a multimeter typically display when testing an open circuit in the resistance mode?
50p
100p
0 Ohms
16
Infinite resistance or OL
8
A specific non-zero resistance value
10
Fluctuating readings
0
Submitted
Oh no, streak lost!
+0

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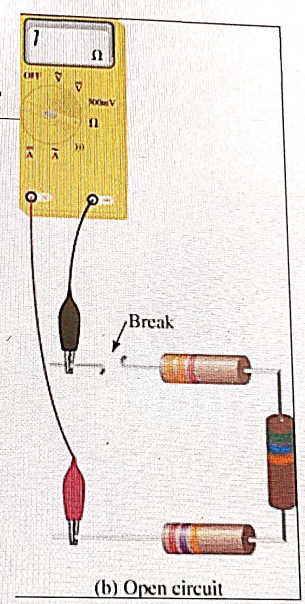
The Open Circuit...



(b) Open circuit

19 10 21 182 34

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Which statement correctly describes the relationship between voltage, current, and resistance in a circuit?

50p 100p

- Higher resistance always increases the current in a circuit. 2
- Current is inversely proportional to voltage but directly proportional to resistance. 7
- Voltage is directly proportional to the product of current and resistance. 14
- Voltage across a circuit component decreases if the resistance is increased while keeping the current constant. 10

Submitted

Incorrect +0

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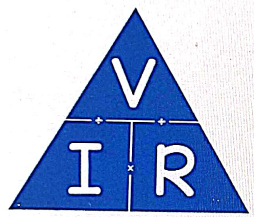
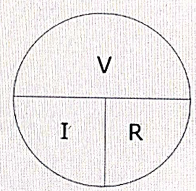
Ohms Triangle



Three Formulas For Ohm's Law

Voltage (V) = Current (I) x Resistance (R)
Current (I) = Voltage (V) ÷ Resistance (R)
Resistance (R) = Voltage (V) ÷ Current (I)

One method to remember formulas based upon Ohm's Law



$$V = IR$$

$$I = \frac{V}{R}$$

$$R = \frac{V}{I}$$

19 10 28 218 43

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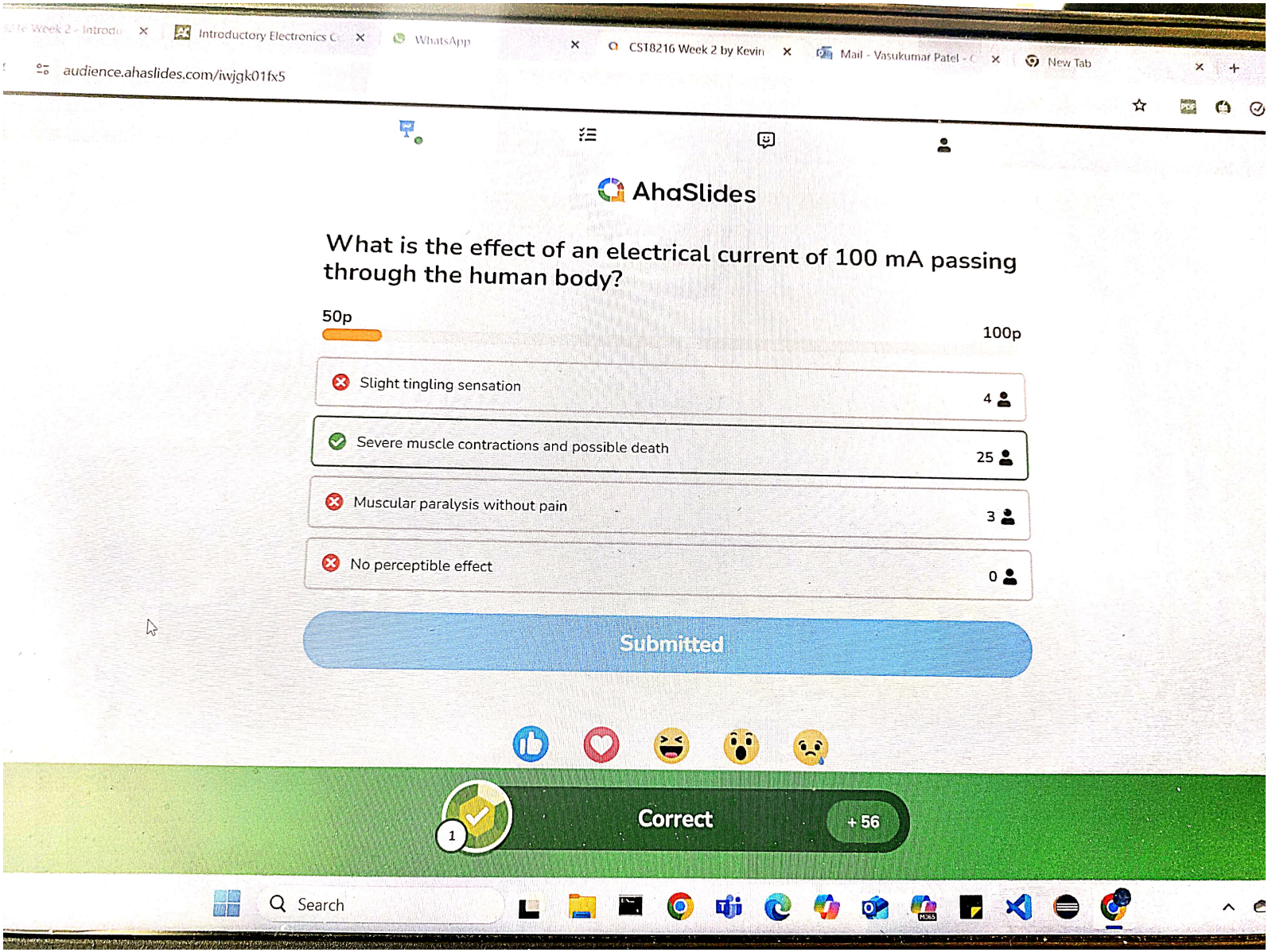
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What is the effect of an electrical current of 100 mA passing through the human body?
50p
100p
Slight tingling sensation
4
Severe muscle contractions and possible death
25
Muscular paralysis without pain
3
No perceptible effect
0
Submitted
1
Correct
+ 56
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Engineering Notation and Metric Prefixes

Power of 10	Prefix	Abbreviation
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

Remember also that $10^{0} = 1$ (unity)

19 10 29 290 43

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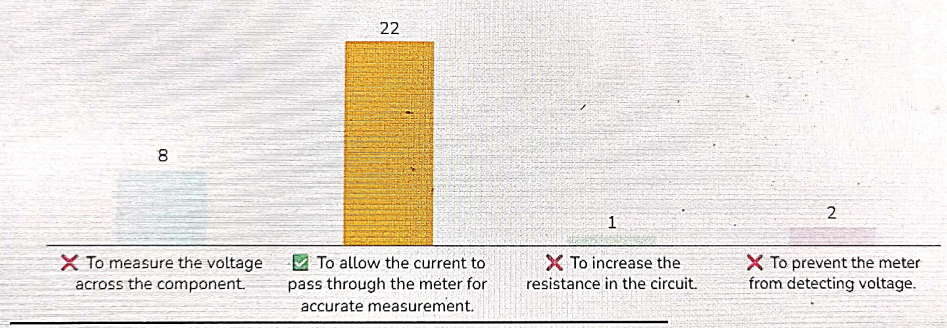
Why must a multimeter be placed in series with the circuit component when measuring current?



27

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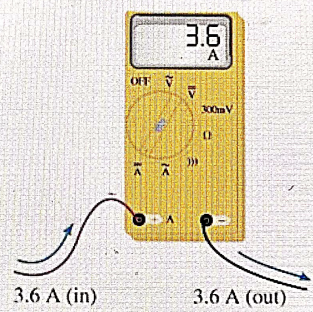
To Measure Current, the current must pass THROUGH the meter



(b) Set selector to $\overline{\overline{A}}$ to measure dc current

28

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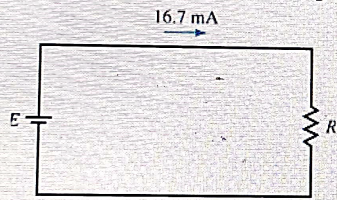
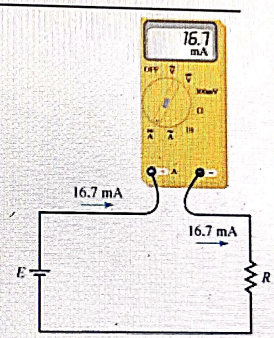
To Measure Current, the current must pass THROUGH the meter – Example Circuit

To Measure Current: Turn off the power, open the circuit, set the meter to DC Amps, then insert the ammeter and turn the power back on again



(a) Current to be measured



(b) Ammeter correctly inserted